

Small-Scale Experiments: Are We There Yet?

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Scaling laws promised cost-effective experiments; six years later, they have yet to fully deliver. Instead, researchers have found them unreliable at small scales (starting at 4M parameters) and concluded that sizable models cannot be avoided. We show this is not the case: the confounding factor is hyperparameters. Small models are highly sensitive, but hyperparameter sensitivity fades with scale. This small-scale sensitivity makes scaling laws easy to miss because they only emerge on the fully tuned frontier, and reaching that frontier requires an extensive search far beyond what most ever run. By ablating the basic scaling law recipe, we show well-tuned hyperparameters matter more than any other ingredient. Further, we reveal why those hyperparameters become easier to find: as scale increases, the hyperparameter loss surface becomes lower dimensional. Nevertheless while scaling laws exist in small models, extrapolation hits statistical limitations. A holistic approach is required. Synthesizing our insights with the recent literature, we develop a new methodology for model-centric research and demonstrate it on a question that once took the field years to settle: where to place normalization layers in the transformer architecture. From small-scale experiments, we recover the large scale result: pre-normalization works better as models grow in size. With the right tools and a better understanding, small-scale experiments can deliver on scaling laws’ long-awaited promise.

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1 Introduction

Foundation models have revolutionized the way we think about computing. They arose as a paradigm only five years ago (Bommasani et al., 2021), yet in that time have become the basis for new discoveries across a broad set of scientific fields (Jumper et al., 2021; Lam et al., 2023; Merchant et al., 2023; Trinh et al., 2024; Hubert et al., 2026). As you might expect, those five years saw a flurry of research into how best to build these models, and one key ingredient emerged: scale. Unfortunately, scale makes experiments incredibly expensive.

The natural solution is to experiment at the small scale, then transfer to the large. Despite the early promise of scaling laws (Kaplan et al., 2020), this solution has not yet materialized after half a decade. Instead, pretraining research grew exponentially in cost, with experiments (not the final training run) dominating the overall expense of model development (Cottier et al., 2025; Denain and Wu, 2026).

Scaling laws failed to deliver because they hit a major roadblock: they are notoriously elusive at small scales. Famously, the two most influential works in the literature reached contradictory conclusions. Kaplan et al. (2020) held parameters should scale much faster than data, as a result early language models were under-trained (Brown et al., 2020; Rae et al., 2022; Chowdhery et al., 2023); Hoffmann et al. (2022) found parameters and data should scale at the same rate. Porian et al. (2024) traced this discrepancy to methodological choices such as whether to count embeddings in the parameters, how long to warm up the learning rate, and scale-dependent hyperparameter tuning—choices that disproportionately affect small-scale models (Pearce and Song, 2024). This early discord foreshadowed later difficulty. Surveying the literature, Li et al. (2025c) found many scaling laws hard to reproduce and, in particular, that scaling laws based on smaller models (100M parameters or less) are unreliable. Li et al. (2025a) aptly summarize the situation as the “*scaling gap* where insights from small-scale experiments often fail to transfer to resource-intensive production systems”.

How do we close this gap and deliver on scaling laws’ original promise? We find that the recipe for a scaling law has one key ingredient that, more than any other, determines final quality: rigorous hyperparameter tuning (§3). By ablating different choices, we show scaling laws are fairly robust to details like how to count

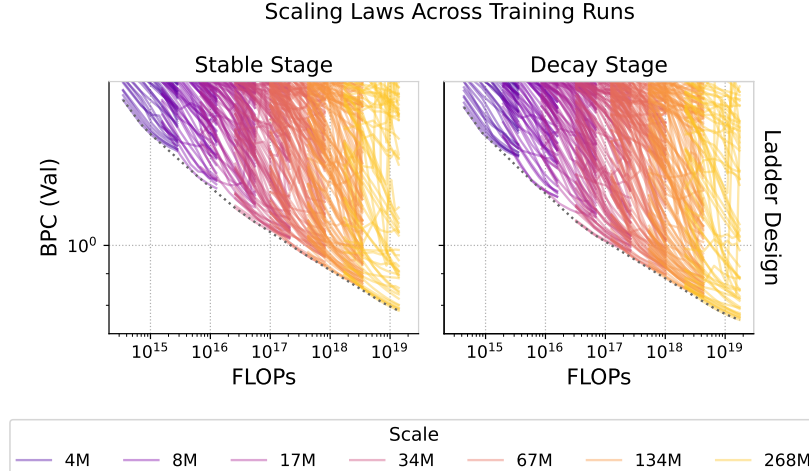


Figure 1 *Scaling laws emerge at very small scales, smaller than widely believed; however, hyperparameter tuning becomes critical. Predictable scaling appears only at the fully tuned frontier. Smaller models are harder to tune, making their scaling laws harder to observe though no less present.*

parameters or whether to decay the learning rate (§3.1). However, poorly tuned hyperparameters can be disastrous (§3.2). Whereas common practice tunes over a small grid, you may need to search hundreds of configurations to reveal the scaling law in small models. Such a search sounds expensive, yet we also find scaling laws extend remarkably far down—as little as 4M parameters, where models train in under an hour on a single GPU, bringing the cost well within reach for many academic and industry labs.

Hyperparameters unduly affect small-scale laws because small-scale models are unduly difficult to tune. We explain this difficulty in terms of the hyperparameter loss surface (§4). As parameters and data grow together, they drive good configurations to fill a larger fraction of the space (§4.1). This proliferation occurs through a geometric mechanism: the intrinsic dimension of the surface goes down (§4.2). The blessing of dimensionality from the parameters cures the curse of dimensionality in the hyperparameters; as model size increases, their *effective* number drops to one. This phenomenon frames how we think about small-scale experiments (§4.3): small scales enable better exploration of the hyperparameter loss surface; large scales make it unnecessary. Since large models are easy to tune, simple rules suffice to carry the hyperparameters up.

Building on these insights, we propose a new methodology for foundation model research drawing on three tools from the literature (§5): the *noisy quadratic limit* reveals if hyperparameters are fully tuned and how hard that is to accomplish (Lourie et al., 2025a); *scaling laws* describe how loss changes as models scale up (Rosenfeld et al., 2020; Kaplan et al., 2020; Hoffmann et al., 2022); and the *perplexity–capability correspondence* answers whether conclusions about pretraining loss have consequences for the capabilities we ultimately care about (Mayilvahanan et al., 2025).¹ Small-scale experiments exploit strong assumptions; the exploration they enable lets us check them. We argue for an approach where each assumption becomes a diagnostic that improves our qualitative understanding of how the model scales (§5.1). This qualitative understanding is important because even though phenomena like scaling laws exist at the small scale, quantitative techniques such as extrapolating the loss compound small errors (§5.2). A holistic approach is required. We demonstrate such an approach on a question that once took the field years to settle (§5.3): where normalization layers should go in the transformer architecture (Vaswani et al., 2017; Baevski and Auli, 2018; Xiong et al., 2020).

In the end, these ideas give a practical methodology for model-centric research:² build a complete understanding at the small scale by thoroughly exploring the model’s hyperparameters, then use simple rules to carry it up. This regimen offers the rare opportunity to cut costs and raise rigor, but it takes the right tools. With them, small-scale experiments can finally deliver on their long-awaited promise.

¹Even when we often can not predict these capabilities in advance (Lourie et al., 2025b).

²*Model-centric* research improves the model holding data fixed; *data-centric* improves the data holding the model fixed. Our discussion focuses on model-centric research while data-centric research likely requires different techniques (§7).

2 Background

Our approach weaves three threads from the literature; we outline each (§2.1–2.3) then our experiments (§2.4).

2.1 The Noisy Quadratic Limit

How do we know if the hyperparameters are fully tuned, or how hard that is to accomplish? The *noisy quadratic limit* answers these questions. [Lourie et al. \(2025a\)](#) found that near the optima, loss surfaces are:

1. **Quadratic**: well approximated by the second-order Taylor expansion;
2. **Low-Rank**: governed by a few directions that explain most of the variation; and
3. **Normal**: distributed about the mean as additive normal noise.

Empirically, this approximation holds for a large region around the optimum, the *asymptotic regime*. This region determines the practical questions we care about, like tuning difficulty or the best attainable performance.

Formally, if $\mathbf{X}$ are the hyperparameters, Y the validation score, and $(\mathbf{x}_*, y_*)$ the optimum, then the conditional distribution $Y \mid \mathbf{X}$ is approximately a quadratic plus additive normal noise, E .³ Letting $H_{\mathbf{x}_*}$ be the Hessian:

$$\mathcal{L}(\mathbf{X}) \approx y_* + (\mathbf{X} - \mathbf{x}_*)^T H_{\mathbf{x}_*} (\mathbf{X} - \mathbf{x}_*) + E, \quad E \sim \mathcal{N}(0, \sigma) \quad (1)$$

This conditional view induces a marginal structure that emerges during random search. The search induces a distribution over validation scores, and that distribution’s tail converges to a *noisy quadratic distribution*, $\mathcal{Q}_{\min}(\alpha, \beta, \gamma, \sigma)$. Writing $F(y; \alpha, \beta, \gamma, \sigma)$ for the CDF of the noisy quadratic distribution, then:

$$\mathbb{P}(Y \leq y) \approx F(y; \alpha, \beta, \gamma, \sigma) \quad \text{as} \quad y \rightarrow y_* \quad (2)$$

The distribution’s parameters capture properties of the loss surface: α is the best attainable performance (y_*), β measures how tightly the scores concentrate around it, γ gives the *effective* number of hyperparameters (intrinsic dimension at the optimum), and σ represents the noise due to random seeds. In practice, we define the asymptotic regime by a threshold on the expected loss, $\mathbb{E}[Y \mid \mathbf{X}] \leq \theta$, marking where this structure holds.

This marginal view makes the limit practical: the high-dimensional surface is hard to estimate, yet we can locate the asymptotic regime with a single scalar and sidestep estimating the Hessian altogether.

2.2 Scaling Laws

Scaling laws classically relate parameters (p) and data (d) to pretraining loss ($\mathcal{L}$). [Hoffmann et al. \(2022\)](#) popularized the most widely used functional form ([Rosenfeld et al., 2020](#)):

$$\mathcal{L}(p, d) = \epsilon + \frac{\zeta}{p^\iota} + \frac{\eta}{d^\kappa} \quad (3)$$

This form omits the hyperparameters, describing loss at the *fully tuned frontier*: the best attainable loss given the parameters and data (Figure 1); however, parameters and data do not vary independently. The model’s gradients must be computed for every example, so their product determines the total compute:

$$c \propto pd \quad (4)$$

Given a budget, you face a trade-off between larger models and longer training. Navigating this trade-off is exactly why people turn to scaling laws. Minimizing Equation 3 with fixed compute yields the condition:

$$\iota \frac{\zeta}{p^\iota} = \kappa \frac{\eta}{d^\kappa} \quad (5)$$

Combining this condition with Equation 4, the optimal parameters and data become power laws in compute:

$$p \propto c^{\frac{\kappa}{\iota + \kappa}} \quad d \propto c^{\frac{\iota}{\iota + \kappa}} \quad (6)$$

³Integral hyperparameters (e.g., batch size) can be treated as continuous; discrete ones are handled with separate quadratics.

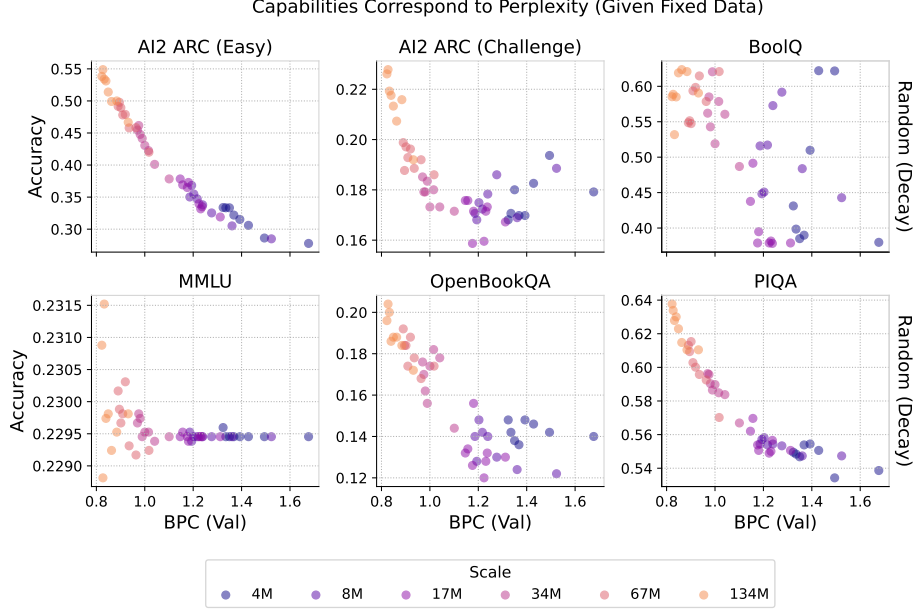


Figure 2 *Given a fixed corpus, perplexity corresponds to downstream capability across a wide variety of tasks. The relationship between pretraining loss and capability is not always predictable or even monotonic; however, models achieving the same loss have a similar conditional distribution over their capabilities as we increase parameters and data, or even randomize the architecture, as long as the *composition* of pretraining data remains fixed.*

If you plug these values into the joint scaling law then you obtain excess loss as a power law in the compute:

$$\mathcal{L}(p, d) - \epsilon \propto c^{-\frac{\iota\kappa}{\iota+\kappa}} \quad (7)$$

Thus scaling laws reveal the best loss given the compute and how to get it by balancing parameters and data.

Hoffmann et al. (2022) found balanced parameters and data grow at the same rate, a conclusion since echoed by many others (Anil et al., 2023; DeepSeek-AI et al., 2024). This idea justifies the token-to-parameter ratio as a meaningful measure of how well-trained a model is, with the compute-optimal ratio serving as baseline. Under Equation 5, this same-rate growth only occurs when the scaling exponents are equal: $\iota = \kappa$. Some theoretical work predicts this equality (Bahri et al., 2024), and the estimates are often close in practice. Researchers have also explored enforcing it to improve estimation (Muennighoff et al., 2025; Li et al., 2025c).

2.3 Perplexity–Capability Correspondence

Scaling laws’ successes in pretraining inspired many to seek them for downstream tasks (Ivgi et al., 2022; Gadre et al., 2025; Chen et al., 2025). Unfortunately, downstream tasks exhibit a wide range of unpredictable behavior, such as emergent (Wei et al., 2022), inverse (McKenzie et al., 2023; Wilcox et al., 2024), or even U-shaped scaling (Wei et al., 2023). Lourie et al. (2025b) found as little as 39% of tasks scale reliably.

Luckily, researchers do not need to predict capability to identify the better model; they only need something that corresponds to it. Remarkably, Mayilvahanan et al. (2025) showed pretraining loss corresponds to capability as long as the pretraining data is held fixed. In other words: models of the same perplexity obtain the same capabilities, even when those capabilities are hard to know in advance. We term this phenomenon *perplexity–capability correspondence*.

The data defines a curve of capabilities; the loss decides a point on that curve; the scaling law determines the cost of that point. Figure 2 illustrates this principle: models trained at different scales or for different lengths fall along a single (possibly noisy) curve. The curve varies wildly across tasks, but exists for all of them. Regardless of whether better capabilities lie down the road, the scaling law sets the toll. In other words a model with a better scaling law achieves every perplexity, and thus any capability, at lower cost.

2.4 Experimental Design

Our experiments explore the hyperparameter loss surface across scales via random search (for details see §A).

Models. We measure model size by the *effective parameter count*, parameters adjusted for FLOPs, so that compute is $c = 6pd$ for p effective parameters and d tokens. This count excludes the embedding layer (which uses no FLOPs) but includes the unembedding layer and the cost of attention (Porian et al., 2024, §B). We target parameter counts at increasing powers of two: $2^{22} \approx 4\text{M}$ to $2^{28} \approx 268\text{M}$, and construct models at each scale with two experimental designs: a hand-crafted model ladder and randomly sampled architectures.

Training and evaluation. We train with a warmup-stable-decay (WSD) schedule (Hu et al., 2024). Following Hägele et al. (2024), we reuse the stable phase from the main run and branch off decay phases from checkpoints taken $1/8, 2/8, \dots, 8/8$ through training. We evaluate checkpoints before and after decaying the learning rate, thus a single sweep yields stable and decayed evaluations across eight token budgets at a fraction of the cost.

Analysis. We partition the scales into training (4M–34M), validation (67M–134M), and test (268M), so reported numbers measure extrapolation from small models to larger held-out ones. By default, we fit the joint scaling law of Equation 3 to the best loss attained at each parameter–data budget pair. Like prior work (Hilton et al., 2023; Li et al., 2025c), we found shorter training runs fit the scaling law poorly, therefore we discard the two smallest token budgets which noticeably improves performance on our validation set.

3 Estimating (Small-Scale) Scaling Laws

What does it take to estimate scaling laws from small-scale experiments? We find them fairly robust to different decisions (§3.1), but one ingredient makes or breaks the resulting law: hyperparameter tuning (§3.2).

3.1 The Impact of Methodology

Despite the frequent debate over parameter definitions, we find they make little difference. Beyond that, we study three refinements on a basic approach: tuning hyperparameters per token budget, decaying the learning rate, and tying the scaling exponents. Some help but none, we will see, is what makes the estimate work.

Parameter definitions only make a small difference. When it comes to scaling laws, researchers define parameters many different ways (Kaplan et al., 2020; Hoffmann et al., 2022; DeepSeek-AI et al., 2024). These definitions differ along three axes: whether to account for attention, the embedding, and the unembedding (Porian et al., 2024, §B). The disagreement pivots on the importance of parameters vs. compute: attention uses more FLOPs on longer contexts, while the embedding layer uses no FLOPs at all. Figure 3 reveals these differences in philosophy lead to little difference in outcome. We use the effective parameters (**Attention** and **Unembedding**, no **Embedding**) as it corresponds most directly to FLOPs per token, keeping $c = 6pd$ exact.

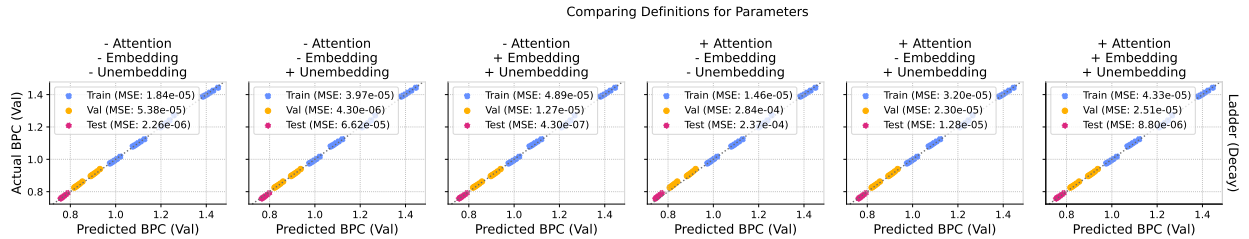


Figure 3 Even at small scales, how you count parameters has limited impact on the scaling law relative to other factors. Researchers count parameters in different ways, diverging on whether to (1) account for attention’s FLOPs (**Attention**), (2) include the embedding (**Embedding**), and (3) include the unembedding (**Unembedding**). The embedding and unembedding have the same number of parameters, giving six distinct counts. Each one yields a reasonable scaling law.

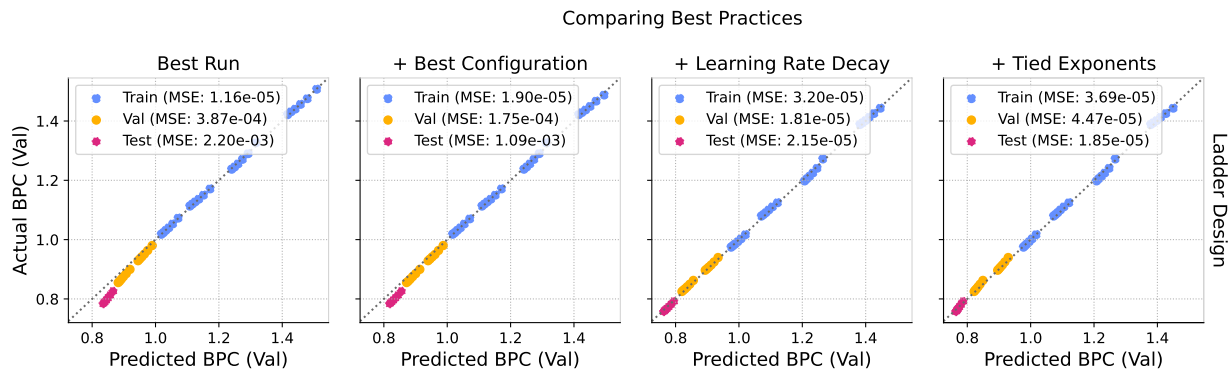


Figure 4 *Better methodology improves the scaling law, but does not make or break it.* A recognizable law emerges from the outset and extrapolation steadily improves as we move from fitting the best run at each scale (left), to the best configuration for each parameter–data budget (middle left), to decaying the learning rate (middle right). Unlike the other refinements, tying the exponents (right) gives inconsistent results: it slightly improves test but hurts validation.

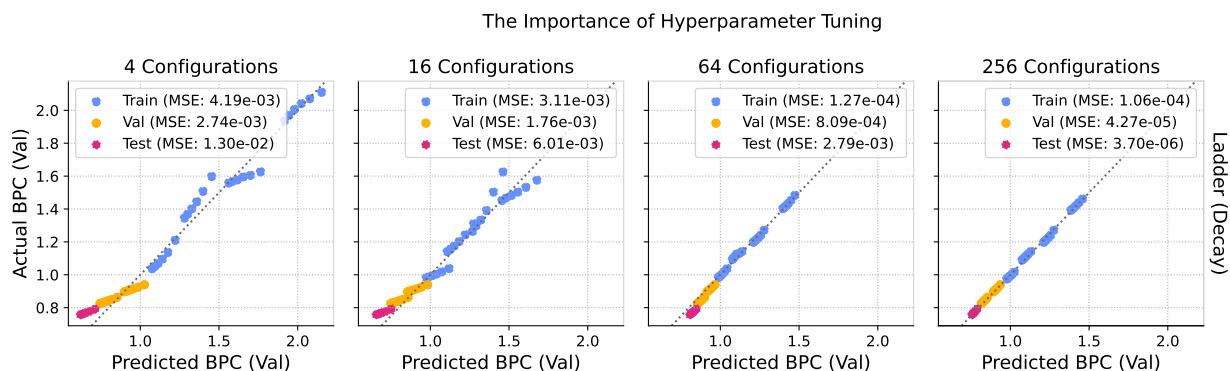


Figure 5 *Hyperparameter tuning is the single most important step in estimating a scaling law.* Each panel fits a scaling law with a more substantial sweep; all panels evaluate (val and test) on the best configurations from our entire sweep. With 4 (left) or 16 configurations (middle left), the law is not even visible. Increasing to 64 (middle right), it clearly emerges but is imprecise. Only at 256 configurations (right) do we obtain an accurate law.

Other choices refine the estimate, rather than enable it. Figure 4 compares increasing refinements on a basic approach. Fitting the best run at each scale without learning rate decay, the scaling law is immediately recognizable (if imprecise). Improving the methodology refines its precision: tuning hyperparameters for each parameter–data budget reduces test MSE by 50%, while decaying the learning rate reduces it by a dramatic 98%. In contrast, tying the exponents ($\iota = \kappa$) makes little difference. The improvements are important for obtaining a good law from small-scale experiments, but even without them the law is present.

3.2 The Importance of Hyperparameter Tuning

Figure 5 shows what happens with too little tuning effort. With 4 configurations per scale, the scaling law is completely absent. With 16, the scaling law has yet to fully appear; it cannot be estimated reliably as the suboptimal configurations inject too much noise. At 64, the law is clearly visible though extrapolation is weak. Only at 256 configurations do we obtain an accurate scaling law. In our previous study, each refinement improved precision on an already recognizable law; here, with too few configurations there is nothing to fit.

Thus one ingredient above all is required to estimate a scaling law, rigorous hyperparameter tuning. This fact explains why small-scale laws are so easy to miss: uncovering them demands a search far more extensive than most ever run. But why are small models so much more sensitive to their hyperparameters? We will show this change in sensitivity comes from how the hyperparameter loss surface evolves as models scale.

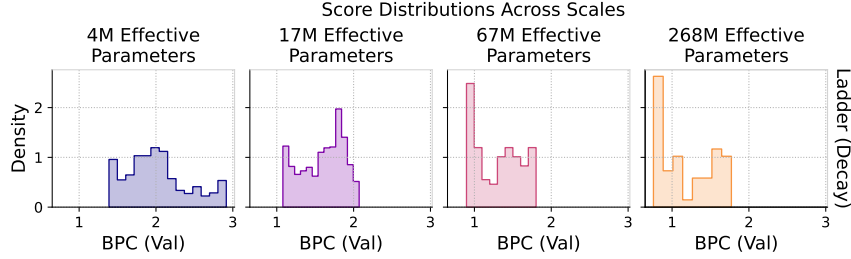


Figure 6 As scale increases, random configurations are more likely to land near the optimum. We measure the volume of near-optimal configurations by visualizing the distribution over validation scores at the end of training from random search (the best 85% to avoid divergent runs). As scale grows, the distribution concentrates near the best achievable loss at each scale, with probability mass piling up into a peak. Good configurations fill the space and get easier to find.

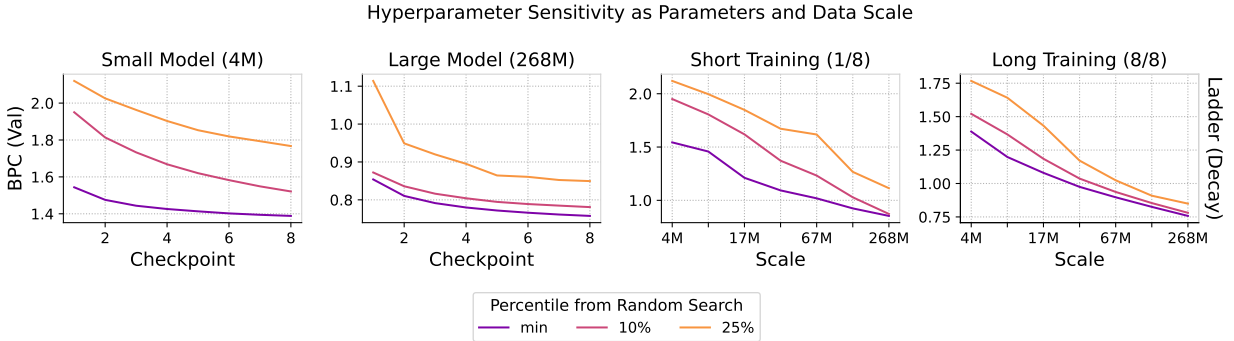


Figure 7 Hyperparameter sensitivity diminishes through increasing parameters and data together. We plot the minimum, 10th, and 25th percentile validation losses from random search. The gap from the minimum measures sensitivity. Scaling data with a small model (left) or parameters with short training (middle right) lowers the loss but only slightly cuts the gap. Scaling data with a large model (middle left) or parameters with longer training (right) collapses the gap.

4 Understanding the Hyperparameter Loss Surface Across Scales

We show that models become less sensitive to hyperparameters as they scale (§4.1). This phenomenon traces back to the hyperparameter loss surface: larger scale makes it lower dimensional (§4.2). These facts influence how we design experiments (§4.3): small scales require extensive search, while large scales are easy to adapt.

4.1 Hyperparameter Tuning Gets Easier with Scale

With scale, good configurations become more common; thus, larger models trained longer are easier to tune.

Near-optimal configurations fill a larger fraction of the space. The effort required to find a good configuration depends on the fraction of the space they take up. We can measure this volume via Monte Carlo integration: sample configurations uniformly and check what percent land near the optimum. This procedure is exactly random search. Figure 6 shows how the score distribution changes with scale: as models grow, the distribution concentrates on the best achievable loss and the density forms a peak. Thus scale makes models easier to tune.

Hyperparameter sensitivity decreases through both parameters and data. Neither parameters nor data alone is responsible for this decrease in sensitivity; the effect comes from both. Figure 7 illustrates how sensitivity decreases across each. It plots the minimum, 10th, and 25th percentile validation losses from random search. The gap from the 25th percentile to the minimum directly measure sensitivity because a small gap means a quarter of all draws are already close to optimal. Scaling one dimension while the other is held small lowers the loss and slightly moves this gap, but scaling one while the other is large collapses it.

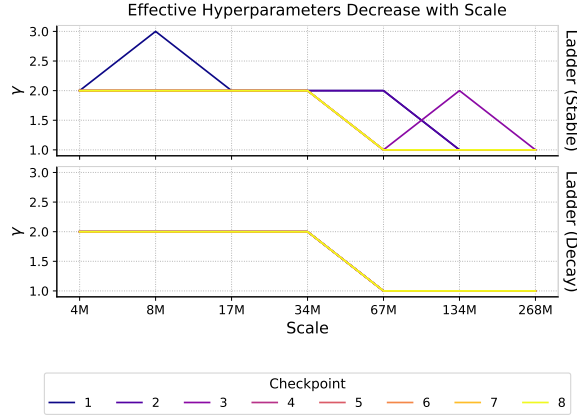


Figure 8 *Big models have fewer effective hyperparameters.* The effective hyperparameter count (γ) is the intrinsic dimension of the hyperparameter loss surface around the optimum. We estimate γ by fitting the noisy quadratic to random search results. γ drops to 1 as models scale.

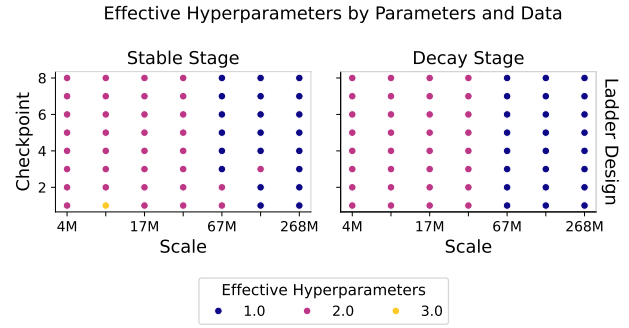


Figure 9 *Parameters, more than data, reduce the effective number of hyperparameters.* The effective hyperparameter count (γ) is the intrinsic dimension of the hyperparameter loss surface. We estimate γ at each parameter–data budget by fitting the noisy quadratic distribution to results from random search. With and without learning rate decay, γ trends lower as parameters increase, while increasing data has much less of an effect.

4.2 The Hyperparameter Loss Surface Becomes Lower Dimensional

The decrease in sensitivity comes from a change in the geometry of the loss surface: it gets lower dimensional.

The effective number of hyperparameters goes down. Intuitively, a model with more hyperparameters is harder to tune; in practice, models can have many while their effective number is still low. Formally, the effective number of hyperparameters (γ) is the surface’s intrinsic dimension at the optimum, a major determinant of tuning difficulty (Lourie et al., 2025a). We estimate γ at each parameter–data budget by fitting a noisy quadratic to the tail from random search (all losses below a threshold, θ). Figure 8 plots these estimates against model size. The trend is clear: larger models have fewer effective hyperparameters, dropping to one.

Parameters are the primary driver of decreasing dimension. Parameters more than data drive the decrease in effective hyperparameters. Figure 9 demonstrates this fact by visualizing γ as a function of both. Checkpoints across training largely share the same intrinsic dimension, while increasing parameters clearly reduces γ .

Thus the effective number of hyperparameters shrinks as parameters grow. Data seems to have less effect. Interestingly, prior work found the effective hyperparameters remains constant when using the same search space across different architectures (Lourie et al., 2025a, §D). It seems counter-intuitive that γ depends on scale but not architecture. Moreover, scale’s effect might change under an alternative parametrization, such as maximum update (Yang et al., 2021). All these factors likely influence γ at some point, our results refine a prior for typical regimes. Better understanding these limits would be a valuable direction for future work.

4.3 Implications for Small-Scale Experiments

The hyperparameter loss surface is not a black box; it has rich qualitative structure. As parameters and data grow, it gets easier to optimize, good configurations fill the space, and the intrinsic dimension goes down.

Can we go beyond qualitative structure to a quantitative model? Unfortunately, nature is generous but not indulgent. We describe several interpretable models in §B (and tried many more). While simple models capture coarse features of the surface, none characterize its finer details. The simplest model, one ignoring how hyperparameters change with scale, is already a strong baseline. This outcome agrees with the qualitative picture: extrapolating to larger scales, hyperparameters have less impact and little variation is left to model.

This picture prescribes where to spend effort. The small scale is higher-dimensional and unforgiving, so the hyperparameters require extensive search; the large scale is gentler and lower-dimensional, so good hyperparameters are easily adapted with standard techniques (DeepSeek-AI et al., 2024).

5 Designing Small-Scale Experiments

We synthesize our discussion into a methodology for small-scale experiments (§5.1). It emphasizes a holistic view because while regularities exist across scales, extrapolation hits statistical limits (§5.2). We demonstrate the approach on a case study from the literature: comparing pre-norm and post-norm transformers (§5.3).

5.1 A Methodology for Small-Scale Experiments

Our approach relies on three facts:

1. with the pretraining data held fixed, capabilities depend on pretraining loss alone (§2.3),
2. with rigorous hyperparameter tuning, scaling laws for pretraining extend to tiny scales (§3),
3. and as models scale, they grow far less sensitive to their hyperparameters (§4).

These principles prescribe a methodology for model-centric research: tune and measure at the small scale, understand how pretraining loss changes, then carry the winner up. The first fact says capability corresponds to perplexity, so we can evaluate models by their cost to obtain it; the second says pretraining laws can be observed at the small scale; and the third says the final scale-up is the easy part.

We expect these assumptions to hold, but should detect when they do not. The shape of small-scale experiments lets us do just that: we can exploit the anticipated structure, check when it fails, and turn those failures into useful diagnostics—for example, if either the noisy quadratic limit or scaling law fails to emerge then there is likely an issue with the hyperparameters, model, or implementation.

More than checking assumptions, however, these diagnostics build a qualitative understanding of how the loss surface scales. Such qualitative understanding is critical because quantitative approaches can compound small errors, even when based on sound structure. More than pure extrapolation, a holistic approach is necessary.

5.2 The Pitfalls of Pure Extrapolation

Though scaling laws exist at small scales, their ability to extrapolate is another matter. We saw small-scale laws successfully extrapolate across an order of magnitude (§3); however, what happens when we go further? Figure 10 illustrates how extrapolation magnifies the sampling error. Comparing independent estimates from different samples, we see the statistical noise: the irreducible error varies wildly across estimates.

If you compare models by extrapolating their scaling laws, at some point you are comparing their estimates of irreducible error. Such estimates are not reliable until the law begins to saturate. Scaling laws exist at small scales, and comparing them is useful; however, you cannot simply pick the model with the lowest final loss. The scaling law is there, but the statistical power is not. Instead, the law is much more reliable near the data.

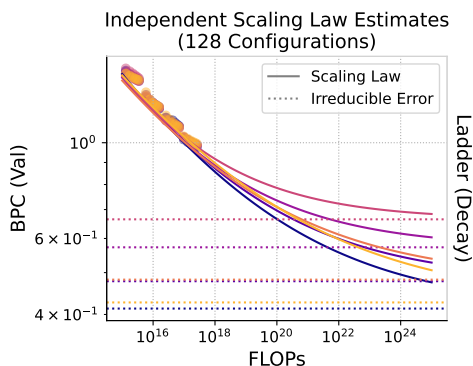


Figure 10 Despite good agreement near the data, extrapolation magnifies small differences due to sampling error. We compare independent scaling law estimates, each based on 128 runs at 4M, 8M, 17M, and 34M parameters. Extrapolation eventually reflects the irreducible error, a quantity whose estimates vary greatly across different samples.

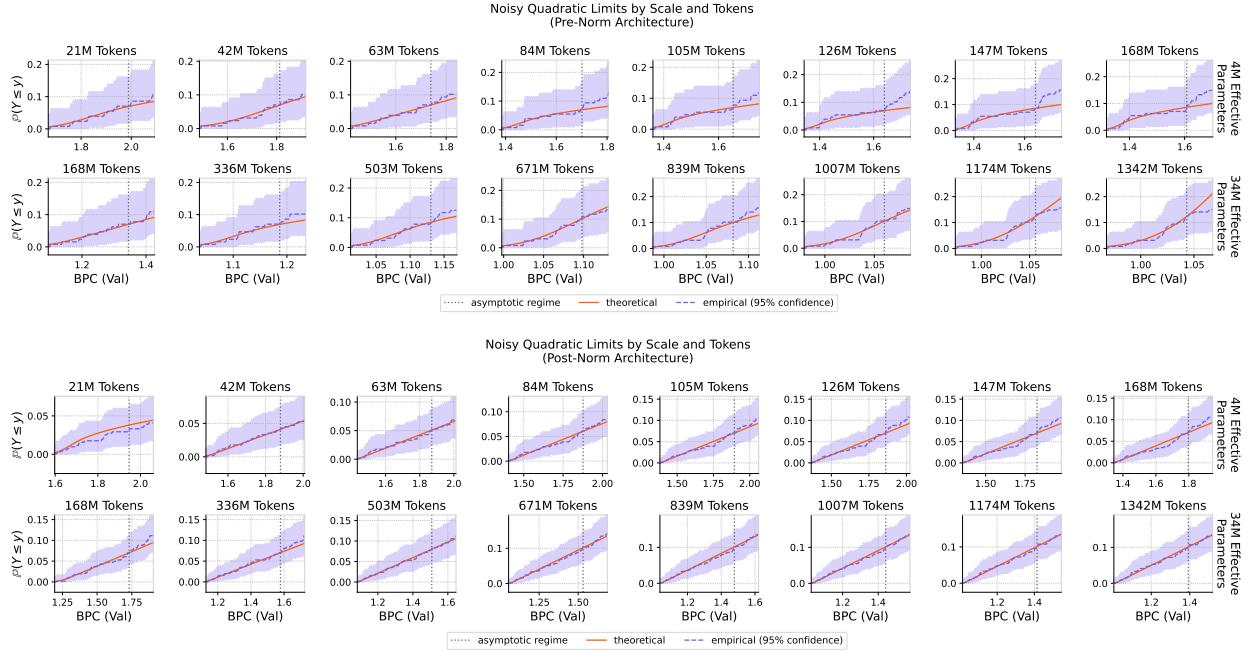


Figure 11 *The noisy quadratic limit emerges for both architectures, confirming the hyperparameters are well-tuned. We fit the noisy quadratic to the distribution of validation losses from random search at 4M and 34M parameters for the pre-norm (top) and post-norm (bottom) architectures. The asymptotic regime is smaller for post-norm than pre-norm, an early sign of its hyperparameter sensitivity.*

5.3 Case Study: Pre-Norm vs. Post-Norm Architectures

Where should normalization layers go in a transformer? The original placed them *after* the residual connection (post-norm) (Vaswani et al., 2017). This made training delicate, needing careful tuning. To stabilize training, Baevski and Auli (2018) moved them *before* the sublayer (pre-norm). A year later, an influential empirical study favored pre-norm in most settings yet, against today’s practice, found post-norm better for high-resource machine translation (Nguyen and Salazar, 2019). The matter only really settled when Xiong et al. (2020) showed theoretically that pre-norm improves gradients at initialization. Pre-norm’s advantage is most pronounced in deeper models, whose experiments are more expensive. Resolving it took nearly three years and a dedicated theoretical analysis. We will recover the answer from straightforward small-scale experiments.

A few well-separated scales should be far more informative than many nearby ones; we use three: two to fit the law, and one to validate it. For post-norm, we sample 511 configurations at 4M, 512 at 34M, and 128 at 134M; for pre-norm, we sample 128 at 4M, 34M, and 134M. We fit on the smaller two then evaluate on the largest one. This search is affordable because when parameters and data scale at the same rate compute is quadratic in model size. A single 1B-parameter run costs as much as 64 runs at 134M, 1,024 runs at 34M, or 65,536 runs at 4M. Thus, our experiment costs as much as a handful of billion parameter runs.

We now deepen our understanding of each architecture through a sequence of qualitative analyses. Each analysis is a diagnostic that asks a question and explores the structure we expect to find.

Diagnostic 1: have we tuned thoroughly? If random search approaches the optimum, then the score distribution converges to a noisy quadratic (§2.1). When the limit fails, it is likely the search missed the optimum.⁴ Figure 11 visualizes the limit for each architecture. For pre-norm it emerges readily; for post-norm it emerges too, but only deeper into the tail. This smaller asymptotic regime is an early indication of increased hyperparameter sensitivity. We seem to have located the optimum, but the tuning difficulty is itself our first finding: post-norm is sensitive to its hyperparameters, as the literature suggests.

⁴The noisy quadratic also enables deeper analyses such as confidence intervals for the best performance (Lourie et al., 2025a).

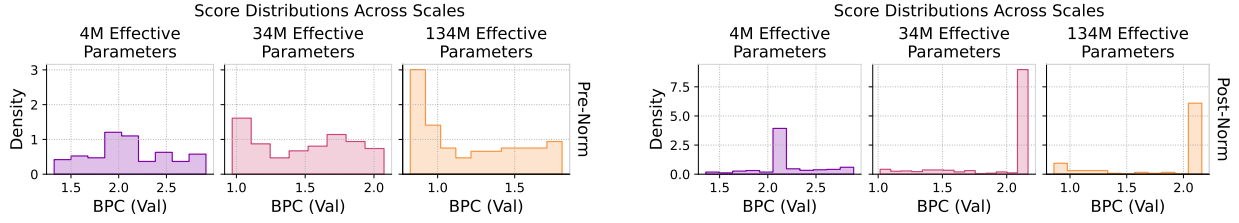


Figure 12 *Hyperparameter sensitivity decreases with scale for both the pre-norm and post-norm architectures.* Across scales, we compare the score distributions (best 85%, to exclude divergent runs) for the pre-norm (left) and post-norm (right) architectures. Both place more probability mass near the optimum as scale increases; post-norm, however, maintains a large peak at a suboptimal value, showing persistent tuning difficulty.

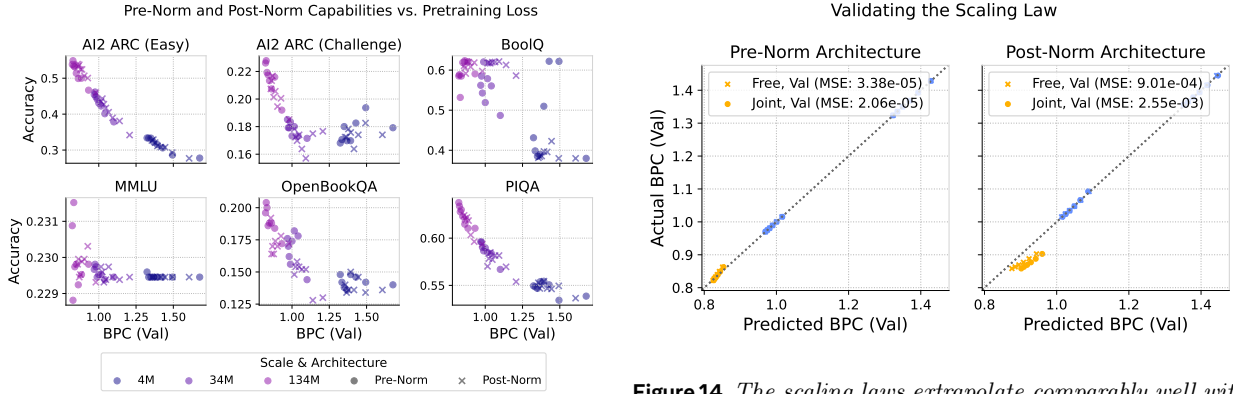


Figure 13 *Pretraining loss corresponds to capabilities both within and across pre-norm and post-norm architectures.* Each panel plots pretraining loss against accuracy for a different task. Across scales and architectures, all points fall on a common trend: equal loss implies equal capability.

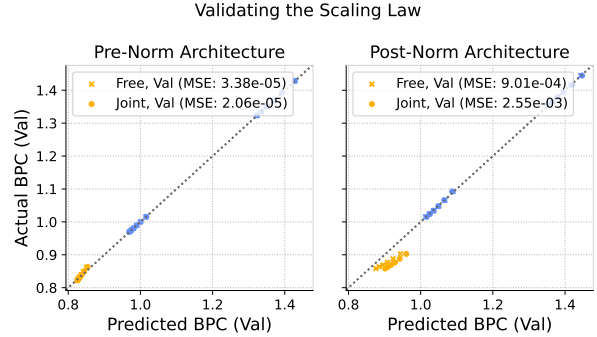


Figure 14 *The scaling laws extrapolate comparably well with free vs. joint irreducible error terms.* We compare sharing the irreducible error term across the architectures (joint) vs. allowing them to differ (free). The pre-norm scaling law extrapolates very well in both cases. The post-norm law extrapolates somewhat, but could likely benefit from even further hyperparameter tuning.

Diagnostic 2: will scaling up be easy? If larger models fail to become less sensitive to their hyperparameters, then transferring from small to large scales might fail as well. Therefore, we examine how sensitivity scales for the architectures at hand. Figure 12 compares the score distribution across scales for each. Both distributions move in the right direction, shifting mass towards the minimum as good configurations fill more of the space. For pre-norm the change is pronounced, growing a distinct peak at the optimum; for post-norm it is only slight, as a second peak lingers at a suboptimal value. Post-norm again exhibits tuning difficulty.

Diagnostic 3: does perplexity still track capabilities? To evaluate by pretraining loss, it must correspond to capabilities (§2.3). This perplexity–capability correspondence should hold when pretraining data is fixed; still, the claim is easy to test as evaluating on a few tasks is much cheaper than pretraining the model. These tasks must show signal at small scales, like AI2 ARC (Easy) (Clark et al., 2018). Figure 13 plots downstream accuracy vs. pretraining loss for several such tasks. Regardless of the architecture, all points fall on a common trend. Equal loss implies similar capabilities and a better pretraining law yields a more cost-effective model.

Diagnostic 4: do the scaling laws emerge? To estimate the scaling laws, we adopt improvements from §3.1: tuning the hyperparameters per parameter–data budget and decaying the learning rate. Since extrapolation primarily reflects the noisy irreducible error (§5.2), we try fitting the laws jointly with tied irreducible error, as well as leaving them free to differ. Figure 14 validates these scaling laws on the held-out scale. The free vs. joint laws extrapolate comparably. Pre-norm exhibits an excellent law, while post-norm’s law is reasonable but could use improvement from further hyperparameter tuning. Again, hyperparameters present a challenge.

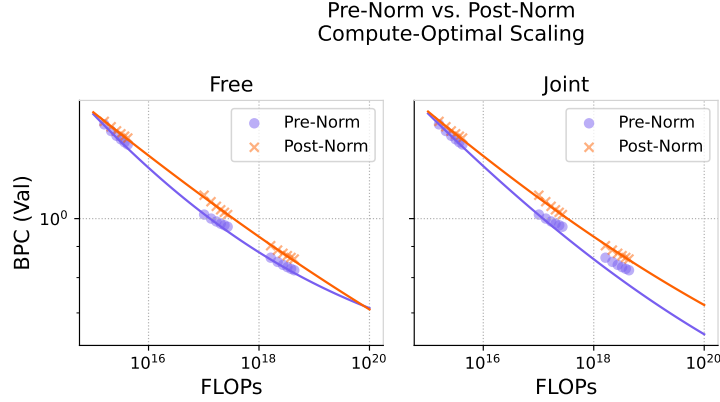


Figure 15 *Pre-norm scales better than post-norm near the data; far from it, extrapolation reflects the irreducible error.* We fit the scaling laws to all available data and compare them under two conditions: separate (left) vs. shared (right) irreducible errors. Across the range we explore, pre-norm scales better. Outside that range, pre-norm continues to scale better if you believe that both architectures saturate at the same final loss. Under this assumption, the gap grows with compute and we recover the large-scale advantage of pre-norm from our small-scale experiments.

With a better understanding of each model, we now pool all sizes together and compare how the architectures scale. Figure 15 presents the result: pre-norm scales better over the range of sizes we explore. Remembering the statistical difficulty of extrapolating the scaling law (§5.2), we consider two kinds of comparisons. If we let each scaling have its own irreducible error, then the post-norm architecture appears to eventually outperform (beyond the scales we tested); however, if we share the irreducible error between the laws then the pre-norm architecture remains more efficient.

From a holistic viewpoint, post-norm was harder to tune at every turn. Even a better architecture in theory is worse in practice if we cannot find good hyperparameters. Beyond that, pre-norm scales better where the scaling laws are most reliable: near the data. Finally, when we assume both architectures reach the same final loss, pre-norm maintains its scaling advantage. These observations support pre-norm as the better choice.

6 Related Work

Researchers have long sought to predict the outcomes of large-scale training from small-scale experiments. Early work studied how models improve with more data (Cortes et al., 1993; Hestness et al., 2017; Sun et al., 2017); later work, with the arrival of web-scale data, turned towards parameters and compute as scaling laws became the primary tool for managing the rising cost of experiments (Rosenfeld et al., 2020; Kaplan et al., 2020). After Kaplan et al. (2020) demonstrated that pretraining loss falls predictably over many orders of magnitude for language models, similar scaling laws were soon discovered in many other domains (Henighan et al., 2020; Jones, 2021; Cherti et al., 2023). Such scaling laws offered an enticing promise: study a model cheaply at the small scale, then extrapolate to the large.

However, that promise proved difficult to keep. Pearce and Song (2024) found small-scale experiments led to discrepancies across major scaling law studies (Kaplan et al., 2020; Hoffmann et al., 2022). More recently, Li et al. (2025c) surveyed the literature, found scaling law estimates hard to reproduce, and reported those built from small models are unreliable. We revisit this regime and reach a more optimistic conclusion: with rigorous tuning, scaling laws exist even at a few million parameters though we must extrapolate them with care.

However, even a flawless pretraining law predicts loss not capabilities, and capabilities have proven far less predictable. Early work sought downstream scaling laws that mirror those found in pretraining: predicting performance directly from parameters, data, and compute (Ivgi et al., 2022; Mahmood et al., 2022; OpenAI, 2023). This work met with mixed success: models matched on these inputs can still differ in pretraining quality, so downstream laws inherit and compound the tuning difficulties of upstream ones—now spread across multiple training stages and a battery of evaluations. To sidestep this issue, researchers recast downstream

performance as a function of pretraining loss itself (Xia et al., 2023; Huang et al., 2024; Du et al., 2024; Gadre et al., 2025; Chen et al., 2025). For some tasks such loss-based laws extrapolate cleanly (Gadre et al., 2025); for many they do not (Lourie et al., 2025b). Many tasks exhibit behaviors that no functional form can extrapolate such as breakthrough (Wei et al., 2022; Srivastava et al., 2023), inverse (McKenzie et al., 2023; Wilcox et al., 2024), or non-monotonic (Wei et al., 2023) scaling. Carefully chosen metrics can mitigate some of this irregularity (Schaeffer et al., 2023, 2025) but not eliminate it (Du et al., 2024; Zhao et al., 2026).

Faced with this irregularity, we must fall back to a more general principle. Instead of predicting capabilities, we rely on a simpler regularity: models with the same pretraining loss tend to have the same capabilities. Researchers first observed this phenomenon across changes in scale (Xia et al., 2023; Snell et al., 2024). The correspondence has since proven remarkably robust. Mayilvahanan et al. (2025) show it survives changes in scale, hyperparameters, architecture, and even the tokenizer. What breaks it is changing the pretraining data, after which the same loss need not imply anything about downstream performance (Lourie et al., 2025b). This perspective lets us rank competing ideas without forecasting their downstream effects because a model family with a better pretraining law reaches every perplexity, and thus any capability, with less compute.

After drawing conclusions at the small scale, you still need the right hyperparameters to scale up. One strategy parametrizes them so that their optimal values stay fixed across scales, as in the maximal update parametrization and its successors (Yang et al., 2021, 2024; Dey et al., 2025; Mlodozieniec et al., 2026). A complementary approach fits scaling laws for the optimal values themselves (DeepSeek-AI et al., 2024; Li et al., 2025b; Bergsma et al., 2025). Many methods work reasonably well because of our structural finding: as models grow, they become less sensitive to their hyperparameters thus a configuration tuned cheaply at the small scale transfers well with just a little adjustment.

We develop several of our insights through the lens of the noisy quadratic limit (Lourie et al., 2025a). Scaling laws and this limit together suggest a simple model of the loss surface across scales might be possible; we find such a unified model elusive but also unnecessary. Closest to this effort, Zhang et al. (2026) predict performance from scale together with the full hyperparameter configuration but their aim differs from ours in spirit: rather than extrapolating from small scales, they fit across scales using publicly available training runs, an approach better suited to modeling familiar configurations than evaluating wholly new ideas.

7 Conclusion

Scaling laws promised experiments would stay cheap; however, this promise has gone unfulfilled. Today, many believe regularity only emerges at large scales, and conclusions drawn at small scales will not transfer.

This misconception stems from a simple, unappreciated fact: small models are much more sensitive to their hyperparameters. While scaling laws extend to very small sizes, they only emerge on the fully tuned frontier. Reaching it requires an extensive search that researchers rarely conduct. Scaling laws predict pretraining loss, and better loss does not always mean better capability; however, loss also fully captures capability—as long as the data is held fixed. Therefore, a researcher who seeks not to predict capabilities but only to improve a model can rely on loss since a model with a better law buys every loss, and thus any capability, at lower cost.

These insights suggest a new methodology of small-scale experiments, one that exploits the strong assumptions we expect to hold true while checking when they do not. Working at small scales makes this rigor affordable: we can train hundreds of models then use statistical tools, like the noisy quadratic limit, to test our assumptions. Small-scale experiments save compute, but more than that they make our conclusions more robust.

Nevertheless, small-scale experiments have several limitations. While scaling laws exist at small scales, their extrapolation hits statistical challenges. The small scale is informative, but requires judgment. Looking at the bigger picture, is the model hard to tune? Is it getting easier? How does scaling compare where we trust the law most? Instead of a single test, we should explore each of these questions using different diagnostics.

Perhaps the most important limitation of all, though, is the restriction to model-centric research. Changing the data breaks the perplexity–capability correspondence which pretraining loss requires to proxy for downstream tasks. With neither pretraining loss as a proxy nor reliable downstream scaling laws, small-scale experiments have limited ability to predict large-scale outcomes. Hopefully future work will unlock small-scale experiments for data-centric research; still, with care we can use small-scale experiments for model-centric research today.

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Appendix

A Experimental Design

This section documents the details and rationales for our experiments.

A.1 Modeling

The modeling recipe we study is a Llama variant, as implemented in Meta Lingua: <https://github.com/facebookresearch/lingua> (commit: 437d680) (Videau et al., 2024). Hyperparameters not specified here were either left to Lingua’s defaults or swept over as described in §A.2.

Data. We pretrain on the 100B token subset of FineWeb-Edu (Penedo et al., 2024). We use the subset to save storage and preprocessing since the scales we consider require less than 100B tokens. We split this subset into training and validation sets using Lingua’s built-in preprocessing pipeline with 48GB of memory and Lingua’s suggested seed of 42. We create two such datasets: one shuffling all the data in 1 chunk for single GPU training (4M–134M parameters), and another splitting the data into 8 chunks for 8 GPU training (268M parameters).

Architecture. For the tokenizer, we use `p50k_base` from `tiktoken` (<https://github.com/openai/tiktoken>), the BPE tokenizer from GPT-3.5 (`text-davinci-003`), because it has an appropriately sized vocabulary for our scale. The context length is 1,024 with special tokens to distinguish the beginning and ending of each sequence. For the neural network, we use a decoder-only Llama model (Grattafiori et al., 2024) available in Lingua (`apps/main/train.py`) (Videau et al., 2024). The model uses RMSNorm for normalization (Zhang and Sennrich, 2019), SwiGLU for the activation function (Shazeer, 2020), and RoPE for position embeddings (Su et al., 2024). All runs use a pre-normalization architecture except for those explicitly specified as post-normalization (e.g., in §5), which we implemented as described by Xiong et al. (2020). Aside from Lingua’s defaults, we padded the vocabulary size (`model.vocab_size`) to 50,536. In addition, we set `model.multiple_of` to 8 since the default value of 256 is larger than our smaller models’ hidden dimensions; 8 compromises between allowing a range of hidden dimensions at small scales while partially exploiting increased GPU efficiency.

Optimization. For optimization, we use AdamW (Loshchilov and Hutter, 2019) with a warmup-stable-decay (WSD) learning rate schedule (Hu et al., 2024). Unlike Lingua, we use a linear cooldown for the WSD schedule decaying to a `optim.lr_min_ratio` of $1e-6$ times the peak learning rate. Each scale we train is defined by a target number of effective parameters (described in §A.2), ranging from $2^{22} \approx 4M$ to $2^{28} \approx 268M$ in powers of two. Training continues until the number of tokens equals 32 times the target number of effective parameters. The total tokens divided by the tokens per gradient step determines the number of steps. The effective parameter count is essentially the same as FLOPs per step as used in Hoffmann et al. (2022) (Approaches 1 & 2), thus we train about 50% further than the Chinchilla-optimal ratio of ~ 20 tokens-per-parameter, allowing us to explore both the under- and over-trained regimes. Following Hägele et al. (2024) and as described in §2.4, we first train each configuration on a main run without decaying the learning rate, saving checkpoints $1/8, 2/8, \dots, 8/8$ through training; afterwards we decay each of those checkpoints by extending training with 25% more steps, thus the final results have their learning rates decayed over the last 20% of training. The initial checkpoints provide our constant learning rate results (*stable*), whereas the decayed checkpoints provide our results for the full schedule (*decay*); all results include learning rate warmup.

Evaluation. For pretraining, we evaluate the model via bits-per-character (BPC), which normalizes the pretraining loss for differences in tokenization. To compute BPC, we change the logarithm’s base on the negative log-likelihood per character already provided by Lingua. We calculate BPC on the validation set, limiting `eval.validation.max_steps` to 1,000. For capabilities, we evaluate the model using Lingua’s integration with the EleutherAI `lm-evaluation-harness` (v0.4.11) (Gao et al., 2023). We evaluate on six multiple-choice tasks: AI2 ARC (Easy) and (Challenge) (Clark et al., 2018), BoolQ (Clark et al., 2019), MMLU (Hendrycks et al., 2021), OpenBookQA (Mihaylov et al., 2018), and PIQA (Bisk et al., 2020). Our evaluations limit `eval.generator.max_tokens` to 16,384 but otherwise use Lingua’s defaults.

```

batch_size ~ DiscreteUniform({64, 128, 256, ..., 4,096})
lr ~ LogUniform(1e-5, 1e-1)
beta1 ~ LogitUniform(0.7, 0.999)
beta2 ~ LogitUniform(0.9, 0.9999)
warmup = round(steps × P), P ~ LogUniform(1e-3, 1/s)
weight_decay ~ LogUniform(1e-4, 1e0)
rope_theta ~ LogUniform(210, 220)

```

Figure 16 The search distribution for all non-architectural hyperparameters. The log uniform and logit uniform distributions are obtained by taking the log or logit of the bounds, sampling uniformly on this scale, then inverting the transformation back to the original scale.

Random Seeds. We fix the following random seeds made available by Lingua: `seed` to 0, `data.seed` to 1, `model.seed` to 2, `eval.harness.random_seed` to 3, `eval.harness.numpy_random_seed` to 4, `eval.harness.torch_random_seed` to 5, `eval.harness.fewshot_random_seed` to 6. Accordingly, data order and other details were kept constant while runs differed primarily in their hyperparameters.

A.2 Search Distributions

Our results analyze three different experiments, each defined by a search distribution over the hyperparameters.

All the experiments randomly sample configurations across a range of scales, with each scale defined by the *effective* number of parameters. The effective number of parameters is proportional to the FLOPs per token so that $c = 6pd$ when c is compute, p the effective parameters, and d the number of tokens. Following Porian et al. (2024, §B), we compute it as:

$$N_{eff} := (3d_{ff} + 4d)dl + dv + ndl \quad (8)$$

where d_{ff} is the feedforward dimension, d the hidden dimension, l the number of layers, v the vocabulary size, and n the context length of the transformer. In particular, parameters in the embedding layer are not counted because they do not contribute FLOPs, while parameters in the unembedding layer are counted because they do. See Porian et al. (2024, §B) for an overview of different ways to count parameters.

To isolate the effect of tuning the architecture, configurations of the same index share all non-architectural hyperparameters across the different designs. Thus, all differences between the experiments come from the architecture alone. The search distribution over the shared hyperparameters is given in Figure 16. Typically, researchers search hyperparameters like the learning rate or weight decay on a log scale; using the proper scale for each hyperparameter is also important in efficiently reaching the noisy quadratic limit (Lourie et al., 2025a). In practice, we found the simple heuristic of sampling real values on a linear scale, positive values on a log scale, and bounded values on a logit scale worked quite well. The warm up is sampled as a proportion of the total training steps up to the first $1/s$ of training so that the first checkpoint (and evaluation) is always fully warmed up.

Model Ladder (ladder). We sample configurations for target scales starting from $N_{eff} = 2^{22} \approx 4M$ up to $2^{28} \approx 268M$ in powers of two. Excluding 2 failed runs, we search 867 configurations at 4M effective parameters, 826 at 8M, 867 at 17M, 934 at 34M, 128 at 67M, 128 at 134M, and 64 at 268M. For the `ladder` experiment, we follow the common practice of setting architectural hyperparameters via a hand-crafted model ladder (Hägele et al., 2024; Bhagia et al., 2025). The hidden dimension, number of layers, and number of attention heads at each scale were set according to Table 1. We increase the width and depth of successive models in order to steadily increase their parameter count, while keeping other architectural elements (such as the `head_dim`) constant. Following Porian et al. (2024), we keep the aspect ratio between 32 and 64. The transformer architecture is subject to a number of constraints, depending on the implementation (e.g., `head_dim` divides

N_{eff}	dim	n_layers	head_dim	n_heads
3.47M	64	2	64	1
7.78M	128	4	64	2
17.1M	256	4	64	4
32.4M	384	6	64	6
62.6M	512	10	64	8
133M	768	12	64	12
270M	1,024	16	64	16

Table 1 The model ladder used for architectural hyperparameters in the *ladder* experiment. The leftmost column gives the actual effective number of parameters which differs slightly from the targeted number due to architectural constraints.

$$\begin{aligned} \text{n_layers} &= \text{round}(U^2), \quad U \sim \text{Uniform}(\sqrt{2}, \sqrt{48}) \\ \text{n_heads} &\sim \text{DiscreteProportional}(\{k \mid k \text{ divides } \text{dim}/8, 1 \leq k < \min(24, \text{dim}/8)\}) \end{aligned}$$

Figure 17 The search distribution for architectural hyperparameters used in the *random*, *prenorm*, and *postnorm* experiments. The discrete proportional distribution samples from a finite set of positive numbers with probability proportional to each number. By construction, `dim` is always divisible by `model.multiple_of` = 8.

`dim`, `dim` is used both for the hidden and the embedding dimension, matrix dimensions should be multiples of 8 when running on a GPU, and so on). These constraints must be considered when designing a ladder.

Random Architecture (random). We sample configurations for target scales starting from $N_{eff} = 2^{22} \approx 4M$ up to $2^{27} \approx 134M$ in powers of two. Excluding 2 failed runs, we search 128 configurations at 4M effective parameters, 126 at 8M, 128 at 17M, 128 at 34M, 128 at 67M, and 128 at 134M. In the *random* experiment, we also tune the architectural hyperparameters via random search. The number of layers and attention heads for each configuration are sampled from the distribution described in Figure 17. Given the effective parameters, the number of layers determines the hidden dimension via Equation 8, which is a quadratic in d given the other parameters and the fact that $d_{ff} = 8/3d$ for our model. We round the solution to the nearest multiple of 8 for GPU efficiency. Once we have the hidden dimension, the number of heads determines the head dimension: $\text{dim} = \text{head_dim} \times \text{n_heads}$. We ensure `n_layers` is at least 2 because induction heads cannot form with fewer layers (Olsson et al., 2022); we also use a square root scale because we found it balanced making both the number of layers and the hidden dimension more uniform. `n_heads` must divide the hidden dimension, so we sample proper divisors of $\text{dim}/8$ to guarantee that `head_dim` is even (required by RoPE) and greater than 16 (required by the attention implementation). Since small divisors are much more common, we bias towards larger ones by sampling divisors with probability proportional to their size. As previously mentioned, transformers are subject to a number of constraints that must be accounted for when sampling architectural hyperparameters: `head_dim` ≥ 16 for common attention implementations, `head_dim` must be even for RoPE, `head_dim` divides `dim`, `dim` provides both the hidden and the embedding dimension, and matrix dimensions should be multiples of 8 for GPU efficiency. These constraints complicate designing a good search distribution;⁵ the distribution in Figure 17 has been constructed to obey them.

Pre-Normalization (prenorm) vs. Post-Normalization (postnorm). We sample configurations for target scales of $N_{eff} = 2^{22} \approx 4M$, $2^{25} \approx 34M$, and $2^{27} \approx 134M$. For *prenorm*, we reuse all the results at those scales from the *random* experiment. For *postnorm*, we reuse the initial configurations of the *random* experiment (including architectural parameters), but we re-run them after changing the transformer to a post-normalization architecture. In addition, we expand the number of configurations due to the post-normalization architecture’s

⁵A typical model ladder avoids most of these complications, as they primarily appear as edge cases in small networks.

increased tuning difficulty. Excluding 1 failed run, this expansion results in 511 configurations at 4M effective parameters, 512 at 34M, and 128 at 134M.

A.3 Analysis

Noisy Quadratic Limit. We generate confidence bands for the eCDF using the LD Highest Density bands from [Lourie et al. \(2024\)](#), and we fit the noisy quadratic distribution as described in [Lourie et al. \(2025a\)](#), both using `opda` (v0.8.0): <https://github.com/nicholaslourie/opda>. To fit the noisy quadratic limit, we set the threshold θ at various percentiles of the score distribution. We used different percentiles for each parameter–data budget, with the size of the asymptotic regime generally increasing in both. In the `ladder` and `random / prenorm` experiments, they were as follows: for 4M, 8M, and 17M effective parameters, 7% at checkpoints 1–6 and 8% at checkpoints 7–8; for 34M, 7% for checkpoints 1–2, 8% at 3, 10% at 4–6, and 12% at 7–8; for 67M, 12% at 1, 15% at 2, 20% at 3, 25% at 4, 30% at 5, 32% at 6, and 35% at 7–8; for 134M and 268M, 12% at 1, 15% at 2, 20% at 3, 25% at 4, 30% at 5, 35% at 6, 40% at 7, and 50% at 8. In the `postnorm` experiment, they were as follows: for 4M effective parameters, 3% at checkpoint 1, 4% at 2, 5% at 3, 6% at 4, and 7% at 5–8; for 34M and 134M, 7% at 1–2, 8% at 3, and 10% at 4–8. We restricted α to be positive (as bits-per-character always is), and we restricted γ to be 1 to at most the nominal number of hyperparameters (7 for `ladder` and 9 for `random / prenorm` and `postnorm`).

Scaling Laws. To fit and evaluate the scaling laws, we partition the scales into training (4M–34M), validation (67M–134M), and testing (268M); thus, we evaluate the scaling laws based on their ability to extrapolate from small to large models. We used training to fit the laws, validation to make decisions during our research, and testing for the final evaluation. For the scaling law, we fit Equation 3 to the best validation loss from each parameter–data budget pair, or (where explicitly stated) to the validation losses from the run achieving the minimum final loss at each scale. In both cases, we drop the first two checkpoints ($1/8$ and $2/8$ of training) as it substantially improved the fit ([Hilton et al., 2023](#)). Where specified, we also tie the scaling exponents together ($\iota = \kappa$). To perform the optimization, we minimized the mean squared error via SciPy’s `scipy.optimize.differential_evolution` ([Virtanen et al., 2020](#)), with bounds of $[0, 1]$ for ϵ , $[0, 10]$ for $\ln \zeta$ and $\ln \eta$ (which we fit on log scales), and $[0, 1]$ for ι and κ . We used a `popsize` of 120 except when fitting laws jointly with a single irreducible error for which we used a `popsize` of 60 due to the higher dimensionality and longer runtime. To evaluate the scaling laws, we report mean squared error between the predicted and observed loss on the held-out scales, with held-out scales using the same preprocessing as those in training (e.g., whether the best run or the best configuration for each parameter–data budget is used).

Interpretable Models of the Hyperparameter Loss Surface. In §B, we compare various models of the hyperparameter loss surface. These models are fit to losses generated via the same methodology used for scaling laws (best loss for each parameter–data budget, dropping the first two checkpoints), except instead of selecting the loss for the single best configuration, we take all configurations within the asymptotic regime as defined by the fractions used for our noisy quadratic limits. Before fitting the models, we transform the hyperparameters to put them on an appropriate scale: a log-scale for `n_effective_parameters`, `n_tokens`, `batch_size`, `lr`, `weight_decay`, `dim`, `n_layers`, `n_heads`, and `rope_theta`; a log-plus-one-scale for `warmup` (to avoid log of zero); and a logit-scale for `beta1` and `beta2`. The interpretable models are fit by first taking a good initialization and then improving it with SciPy’s `scipy.optimize.minimize` using the "Nelder-Mead", "Powell", and "BFGS" methods in sequence. For the initialization, we fit a quadratic polynomial via ordinary least squares, algebraically solve for the equivalent estimate in the format of Equation 10, and then project the Hessian to a lower rank ($\gamma = 3$) by dropping the eigenvalues of least absolute value. We then use the `static` model to initialize the `dynamic optima` and the `dynamic sensitivity` models, setting the new components to zero since the models are nested. We similarly initialize the `dynamic` model from the `dynamic optima` model. For the `multilayer perceptron` model, we implement the neural network using scikit-learn ([Pedregosa et al., 2011](#)). The pipeline applies a quadratic basis expansion to the hyperparameters, normalizes the features via `preprocessing.RobustScaler`, and finally predicts via a multilayer perceptron (MLP) with `hidden_layer_sizes=[128]`, `learning_rate="adaptive"`, `batch_size=256`, and `max_iter=4096`. We grid search `alpha` over `1e-4`, `1e-3`, `1e-2`, `1e-1`, `1e0` and `learning_rate_init` over `1e-5`, `3e-5`, `1e-4`, `3e-4`, `1e-3` using the train-validation split described previously. For the MLP, we fit the scaling law beforehand and then fit the MLP to the percent excess loss, rather than fitting them jointly.

A.4 Implementation

Our implementation further involves the following details.⁶

Training Pipeline. We use gradient accumulation so that we can tune the batch size despite different scales requiring different amounts of memory and GPU setups (1 vs. 8 GPUs). We set the per-GPU batch size to the largest power of two that safely avoids out-of-memory errors in order to maximize throughput while also evenly dividing the number of tokens (16 sequences for 4M–67M parameters, 8 sequences for 134M–268M parameters). Each scale is assigned a base number of gradient accumulation steps chosen so that after accounting for the per-GPU batch size and the number of GPUs the model will process 64 sequences per update. We then multiply that base by a power of two to achieve the distribution described in Figure 16 for the total batch size. As a result, the batch size sweep is hardware-agnostic and always evenly divides the total number of training tokens. In addition, we set `gc_collect_freq` to 1,024, `data_prefetch_size` to 1,024, `distributed.compile` to `true`, and `distributed.fsdp_type` to `"full_shard"` in order to make training efficient. Finally, we set `logging.freq` to 1 and found it necessary to set `env.ENABLE_INTRA_NODE_COMM` to `"0"` to avoid an unresolved error in the PyTorch version.

Software. We train the models using Meta Lingua (commit: 437d680), as previously described (Videau et al., 2024). We modify the code base to support the 100B token subset of FineWeb-Edu, use a linear decay in the WSD schedule, and use PyTorch’s Scaled Dot Product Attention (SDPA) instead of Flex Attention as Flex Attention would not successfully generate kernels for some of the transformer shapes that we randomly sample. In addition, we switch the transformer architecture from pre-norm to post-norm for some experiments as previously discussed. We isolate the software environment in an Apptainer (Singularity) container (Developers, 2021). The container runs Ubuntu 24.04.2 LTS with CUDA 12.8 (V12.8.93) and cuDNN 9.8, and we run Lingua via a Python 3.11.14 conda environment with PyTorch 2.7.0+cu128 (Paszke et al., 2019).

Hardware. Training was done on 80GB NVIDIA A100 GPUs on a SLURM cluster. Model sizes from 4M–134M effective parameters were trained with 1 GPU, while models with 268M effective parameters were trained data-parallel with 8 GPUs.

B Interpretable Models of the Hyperparameter Loss Surface

In §4.3, we discussed developing an interpretable model of the hyperparameter loss surface across scales by combining the noisy quadratic limit with scaling laws. We reported that such models capture the broad shape of the surface, but not precisely how it evolves. Here we provide those explorations in more depth. In particular, we will compare several approaches a researcher might be tempted to try and show that while they have some success in modeling the loss surface, they are far from the clear wins available via scaling laws or the noisy quadratic limit on their own.

Scaling laws model the optimal loss across scales; the noisy quadratic limit models how hyperparameters influence the loss at one scale. To combine them, we must set the quadratic’s minimum according to the scaling law. One could accomplish this in several ways;⁷ a simple approach with a strong intuition starts from the observation that, given any reasonable hyperparameters, the loss should approach the irreducible error (ϵ) as parameters and data go to infinity. We capture this by modeling the percent increase in *reducible* loss at each scale as a quadratic in the hyperparameters:

$$\mathcal{L}(p, d) \approx \epsilon + \left(\frac{\eta}{d^\kappa} + \frac{\zeta}{p^\iota} \right) \left(1 + L\mathbf{X} + (\mathbf{X} - \mathbf{x}_*)^T H_{\mathbf{x}_*} (\mathbf{X} - \mathbf{x}_*) \right) \quad (9)$$

Here, L represents residual linear terms necessary to make any quadratic expressible.⁸ With this model, the

⁶Preliminary experiments were run on the NYU Torch cluster. We later expanded our random searches using Meta’s compute infrastructure. This section describes the compute setup used at NYU, which is sufficient to reproduce our findings.

⁷Beyond the approach discussed here, we also tried using the scaling law for the quadratic’s constant term which is perhaps the most obvious approach even if it does not have an a priori justification.

⁸One might expect $L = \mathbf{0}$ at the optimum. While we do not enforce this assumption in our experiments, we explored it and the conclusions are similar.

	Validation (MSE)		Test (MSE)
	67M	134M	268M
Scaling Law	0.000011	0.000025	0.000020
Parameters and Tokens Only	0.011471	0.014014	0.019795
Polynomial	0.004452	0.005217	0.008790
Static	0.004820	0.005900	0.010027
Dynamic Optima	<i>0.002824</i>	<i>0.004401</i>	<i>0.006630</i>
Dynamic Sensitivity	0.005413	0.007185	0.010813
Dynamic	0.003672	0.006806	0.010102
Multilayer Perceptron	0.008866	0.012003	0.015464

Table 2 The mean-squared error (MSE) of different models for the hyperparameter loss surface extrapolating from small to large scales using our model ladder and decaying the learning rate. The scaling law predicts loss at the optimal hyperparameters; the rest predict all configurations in the asymptotic regime. Each column’s best loss is italicized (excluding the scaling law). The static quadratic that ignores scale is already a strong baseline; varying the optimum appears to help while varying the curvature seems to hurt. The **dynamic optima** model performs best here; however, the models’ ranking is sensitive to optimization details like the initialization. Every variant improves over predicting from parameters and tokens alone (ignoring the hyperparameters), but none reaches the error attained by the scaling law when predicting the loss at the optimal hyperparameters; in other words, no model fully captures the surface.

scaling law supplies the optimal loss across scales while the quadratic inflates the reducible part according to how far the hyperparameters sit from the optimum.

Since the Hessian is real and symmetric, we can diagonalize it: $H_{\mathbf{x}_*} = UDU^T$. Let Λ be the diagonal matrix of non-zero eigenvalues and take $M_{\mathbf{x}_*}$ to be the submatrix of U^T that projects onto their corresponding eigenvectors. Letting $\mathbf{z}_* = M_{\mathbf{x}_*}\mathbf{x}_*$ represent the linear subspace of optimal hyperparameters, we can write this model as a quadratic in the (lower-dimensional) projected space:

$$\mathcal{L}(p, d) \approx \epsilon + \left(\frac{\eta}{d^\kappa} + \frac{\zeta}{p^\iota} \right) \left(1 + L\mathbf{X} + (M_{\mathbf{x}_*}\mathbf{X} - \mathbf{z}_*)^T \Lambda (M_{\mathbf{x}_*}\mathbf{X} - \mathbf{z}_*) \right) \quad (10)$$

This form enables us to fit the model with a low rank constraint.

We explored several variants of this model, in all cases restricting the Hessian’s rank to at most $\gamma = 3$. The **static** model fixes both the optimum and the curvature across scales as in Equation 10. The **dynamic optima** model keeps the curvature fixed but makes the (latent) optimal hyperparameters a linear function of the log scale: $\mathbf{z}_* = \mathbf{z}_0 + \log(p)\mathbf{z}_p + \log(d)\mathbf{z}_d$. The **dynamic sensitivity** model instead fixes the optimum and lets the curvature vary, making the eigenvalues of the Hessian power laws of the scale: $\lambda_i p^{\nu_i} d^{\xi_i}$. The **dynamic** model varies both. For comparison, we include a **polynomial** model which predicts the percent increase in excess loss using a quadratic polynomial (with no constraint on the rank). We also compare against a **multilayer perceptron** model that abandons interpretability altogether and predicts percent increase in excess loss as a nonlinear function (neural network) of the hyperparameters and scale.

Table 2 compares these variants, fit on the small scales and evaluated on the larger held-out ones via mean-squared error (MSE). Ignoring the scale’s effect on the hyperparameters (**static**) already performs surprisingly well. The dynamic variants extrapolate with mixed success: varying the optimum appears to help, while varying the curvature seems to hurt. The nonlinear model is the most flexible; here it performs the worst of all models that incorporate the hyperparameters, though it can also perform the best in similar setups depending on optimization details like the initialization. While every variant beats the baseline of ignoring the hyperparameters and predicting from the parameters and tokens alone, none fully captures the loss surface. If one did, it would approach the much smaller error achieved by the scaling law when predicting the optimal hyperparameters.